THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** the container and the flame also differ along with the volume	Volume of water is a variable.
2	**(b)** a grass	Grasses show these features.
3	**(c)** starch and fat	Both tests are positive.
4	**(d)** P and R are magnets, Q is magnetic material	Repulsion is seen only between magnets.
5	**(c)** 160 cm	320 - 160.
6	**(c)** P	The pin falls first.
7	**(c)** 50 °C	Linear scale with equal spacing gives 50 °C.
8	**(d)** heat is used to change the state	Latent heat idea.
9	**(c)** magnet, add water and filter, evaporate filtrate	Each step removes one component.
10	**(a)** it does not grow or reproduce on its own	Many characteristics are absent.
11	**(d)** most oxygen was used	Burning consumed oxygen.
12	**(d)** 8 light-minutes	Approx.
13	**(a)** both depend on hot weather	A third factor links them.
14	**(c)** helps judge distance for hunting	Forward eyes give depth perception.
15	**(a)** they contain much sugar and few nutrients	Empty calories.
16	**(a)** iron is magnetic and copper is not	Only iron is attracted.
17	**(a)** time period is 1.8 s in both	T = 36/20 = 1.8 and 18/10 = 1.8.
18	**(c)** B	Translucent materials.
19	**(a)** 40 s	Twice slower.
20	**(b)** 100 g	Mass is conserved.
21	**(d)** 30 g	Same as initial.
22	**(a)** pebble	It has one cell.
23	**(b)** adding compost	Compost adds humus.
24	**(d)** 60 kg	Mass does not change.
25	**(b)** compare the methods and repeat carefully	Differences in method may explain.
26	**(c)** many plants will fail to pollinate	Pollination would fall.
27	**(a)** fat	Protein test.
28	**(d)** always in the same direction with the same pole of a magnet	Opposite pole forms at the lifting end.
29	**(d)** 10.5 cm	12.5 - 2.
30	**(c)** its shape displaces much water	Hollow shape lowers average density.
31	**(c)** aluminium and plastic	Pan conducts, handle insulates.
32	**(a)** it changed to vapour and mixed with air	Matter is conserved.
33	**(d)** it makes tiny particles clump and settle	Coagulation.
34	**(b)** a living thing can stay inactive until conditions suit it	Life processes were only slowed down.
35	**(a)** more public transport	Fewer vehicles.
36	**(a)** Earth rotates on its axis	Tilt.
37	**(a)** more than one variable was changed	Only one variable should vary.
38	**(c)** frog	Frogs show this.
39	**(d)** Q contains starch	Iodine tests starch only.

40	**(a)** south-seeking pole	The end that points south is the south-seeking pole.
41	**(a)** Both A and R are true and R is the correct explanation of A.	A hypothesis is a testable explanation, and testing it by an experiment is exactly how we check it.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R is a general pattern and does not explain why wheat has fibrous roots.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Growth and repair of tissues is why proteins are called body-building nutrients.
44	**(a)** Both A and R are true and R is the correct explanation of A.	The Earth's magnetic field acts on the suspended magnet and aligns it, so R explains A.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the conversion does not explain why metre is the SI unit.
46	**(a)** Both A and R are true and R is the correct explanation of A.	We can see clearly through glass because light passes through it; R explains A.
47	**(a)** Both A and R are true and R is the correct explanation of A.	After holding a hand in cold water, lukewarm water feels warm; this shows touch can mislead, which R explains.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed boiling temperature does not explain why evaporation occurs at every temperature.
49	**(a)** Both A and R are true and R is the correct explanation of A.	The difference in weight is used with wind, so R explains A.
50	**(a)** Both A and R are true and R is the correct explanation of A.	Not showing life processes is why a stone is non-living.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** protein and fat	Plant foods have less protein and fat.
2	**(b)** iron only	Only iron is magnetic.
3	**(d)** it is in linear motion and its speed increases	It falls along a straight line, speeding up.
4	**(a)** C conducts heat and D does not	Test for conduction.
5	**(b)** the 20 °C object	Heat enters the colder one.
6	**(a)** 10 mL	50/5.
7	**(d)** 18 g	Half.
8	**(c)** water is needed for germination	Controlled comparison.
9	**(a)** loosening and mixing it	Aeration.
10	**(a)** one sixth	Weaker gravity.
11	**(b)** rounding off numbers to look neat without noting it	Data must not be altered.
12	**(d)** it provides more food, water and shelter	Rich resources support life.
13	**(a)** X has starch, Y has fat, Z has neither starch nor fat	Match tests with nutrients.
14	**(c)** B's north repels A's north	Like poles repel.
15	**(b)** 8.3 cm	All readings are higher.
16	**(a)** it is light and cushions shocks	Soft and light.
17	**(a)** metal rod hotter	Metal conducts quickly.
18	**(c)** smaller volume and large surface to depth ratio	Less water.
19	**(c)** filtration cannot remove all germs	Disinfection is needed.
20	**(c)** change in body during growth	Life cycle.
21	**(b)** planting a tree for every tree cut	Balance.
22	**(b)** stars appear to move across the night sky	Apparent motion.
23	**(a)** revise the hypothesis	Data must be respected.
24	**(c)** to avoid sinking in sand	Large area reduces pressure.
25	**(b)** spinach and jaggery	Iron-rich foods.
26	**(b)** repel	N facing N.
27	**(b)** 1200 m	2.5 x 480.
28	**(d)** iron is heavier for the same size	Comparing same volume.
29	**(d)** heat in equals heat out	Balance.
30	**(d)** the colour of the container	Not relevant.
31	**(d)** fine clay	Smaller particles settle slowest.
32	**(b)** it swells and sprouts in water	Growth suggests life.
33	**(d)** nitrogen > oxygen > carbon dioxide	78, 21, 0.04.
34	**(b)** 30 March	About 29.5 days.
35	**(a)** growth of the plant	Growth is the dependent variable.
36	**(b)** being active at night	Avoid heat.
37	**(c)** cereals with pulses	Combination improves protein quality.
38	**(b)** Earth's magnetism slowly magnetises it	Induction by Earth's field.
39	**(d)** a rolling ball	Rolling ball.

40	**(b)** using a magnet	Magnet only affects iron; keeps others intact.
41	**(a)** Both A and R are true and R is the correct explanation of A.	Because it is continuously available, sunlight is renewable.
42	**(a)** Both A and R are true and R is the correct explanation of A.	A star makes its own light and heat, so R explains why the Sun is a star.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the use of a telescope does not explain why observation comes first.
44	**(a)** Both A and R are true and R is the correct explanation of A.	The smooth, tapering body reduces resistance in water, so R explains A.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the food source does not explain the disease.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R does not explain why iron is attracted.
47	**(a)** Both A and R are true and R is the correct explanation of A.	Non-standard units give different values for different people, so a common standard unit is needed.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are about different materials; R does not explain A.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the two uses are separate and R does not explain A.
50	**(a)** Both A and R are true and R is the correct explanation of A.	Wind removes vapour near the cloth and speeds up evaporation.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** to reduce random errors	Average.
2	**(c)** it may rust or conduct heat	Metals conduct heat and may rust.
3	**(a)** electric iron handle	Handle plastic.
4	**(a)** melting of ice	Water can be refrozen.
5	**(a)** sedimentation, filtration, disinfection	Standard.
6	**(c)** accumulated wastes are toxic	Removal of harmful substances.
7	**(b)** no fuel cost and no smoke	Clean.
8	**(a)** halve	12 hours.
9	**(b)** Does temperature affect how fast sugar dissolves?	Testable questions have measurable answers.
10	**(c)** thick fur, fat layer, white colour	Cold adaptation.
11	**(b)** vitamin A, night blindness	Vitamin A deficiency affects dim-light vision.
12	**(d)** a magnet or iron object nearby	Local magnetic influence.
13	**(b)** 47 cm	188/4.
14	**(a)** sugar and chalk powder both look white but behave differently in water	One dissolves, the other does not.
15	**(c)** the water is at the same temperature as the room	Same as surroundings.
16	**(a)** increase	Larger area.
17	**(c)** decantation after the layers form	Layers form.
18	**(d)** movement	Machines and rivers move.
19	**(d)** less soil washed away	Conservation.
20	**(c)** sunset	Opposite the Sun.
21	**(a)** curiosity about everyday things can lead to great discoveries	Simple tools with careful thinking can lead to discoveries.
22	**(a)** prop roots	They support the tree.
23	**(c)** butter and dal	Butter has fat; dal protein.
24	**(d)** the pin acts as a temporary magnet	Induced magnetism.
25	**(d)** each blade moves in a circular path	Blades go round.
26	**(a)** sand on the filter paper	Sand is insoluble.
27	**(a)** 20 °C	Below its range.
28	**(b)** 180 mL	3600/20.
29	**(a)** 20 g	Half of 40.
30	**(b)** growth from within using food	Cell division.
31	**(a)** nutrient loss in soil	Monoculture.
32	**(a)** 90 minutes	42000/90.
33	**(d)** the plant grows with time	Rising line indicates growth.
34	**(d)** to store water	Stem stores water.
35	**(c)** include pulses, seasonal vegetables and local fruits	Local foods are cheap and nutritious.
36	**(a)** flat on the palm, away from iron objects	Keep it level and away from magnets.
37	**(d)** placed touching the object and viewed straight	Reduce gap.
38	**(d)** heating suddenly	Sudden heat can crack it.

39	**(a)** how hot something is	Hotness.
40	**(b)** it increases	More evaporation.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are different mixtures and methods; R does not explain A.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they describe different organisms; R does not explain A.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are different topics and R does not explain A.
44	**(a)** Both A and R are true and R is the correct explanation of A.	The Moon only reflects the Sun's light.
45	**(c)** A is true but R is false.	A is true because repeating avoids mistakes; R is false, a single trial can be wrong.
46	**(c)** A is true but R is false.	A is true; R is false because cactus leaves are small spines.
47	**(c)** A is true but R is false.	A is true (starch test); R is false, the blue-black colour is the test for starch only.
48	**(d)** A is false but R is true.	A is false: like poles repel. R is true.
49	**(a)** Both A and R are true and R is the correct explanation of A.	Repeating after equal time intervals is the meaning of periodic motion.
50	**(d)** A is false but R is true.	A is false: sand is insoluble. R is true.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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